11 Conclusion

This paper is intended as a reference catalog and an argument for continuous valued market based metrics. Its central claim is that replacing discrete authorship with continuous contribution shares, and unweighted citations with normalized and corrected weighted ones, unlocks more accurate and insightful scientometrics, while a coupled academic marketplace for trading credit for services unlocks more reliable and robust academic quality control.

Central to the construction of Liberata's scientometrics are the shares graph G_S and the references graph G_W, which together compose the capital graph $G_{\divideontimes}$ and support a broad and deep set of capabilities. Graph-theoretic properties of the shares network, including its spectral structure, spanning tree counts, and two-step compositions, characterize collaboration topology, global connectivity, and potential collusion between authors and their quality controllers.

Portfolio metrics, well defined over any collection of manuscripts and contributors, provide measures of impact, concentration, collaboration structure, and research efficiency that extend naturally from individuals, to institutions, fields, and geographic regions resolving attribution ambiguities that render noisy comparisons in the current system.

Market metrics derived from share transactions yield field specific value of quality control services, risk premiums of entities, and risk-adjusted performance measures that have no analogy in conventional metrics. Modular correction factors to the references graph, linearly composable, provide a flexible system for comparing and counting how impact propagates through the citation network under different impact definitions.

Liberata's publishing system offers a way to correct incentive alignment problems between authors, quality controllers, and the broader academic community. Liberata's scientometrics offers a way to detect each type of conventionally recognized and exercised exploitative strategy, enabling such a platform and many other rich scientometric applications to be developed in future work.

12 Notation

12.1 Node Classes

$m \in \hat{M} \subset M$	Manuscripts
$a \in \hat{A} \subset A$	Authors
$p \in \hat{P} \subset P$	Peer reviewers
$r \in \hat{R} \subset R$	Replicators
$c \in \hat{C} \subset C$	Contributors
$i \in \hat{I} \subset I$	Institutions
$t \in \hat{T} \subset T$	Timestamps
$\theta \in \hat{\Theta} \subset \Theta$	Geographic regions

12.2 Intermediates

x, y, z	Various indexing
e	Euler's constant
λ, Λ	Eigenvalue, Diagonal Matrix of Eigenvalues
$k \in K, l \in L, n \in N$	Counters
$\mathbb{R}$	Real numbers
$\mathbb{N}$	Natural numbers
$\mathbb{Z}$	Integers
$\mathbb{E}$	Expected value
b	Arbitrary parameter
$\bar{s}$	Bar indicates vector of the object type underneath the bar

12.3 Graphs & Matrices

$u \in \hat{U} \subset U$	Unweighted citations
$w \in \hat{W} \subset W$	Weighted citations
G	Graph
$\mathbf{G}$	Adjacency matrix of graph
$\mathbf{B}$	Block of a matrix
V	Set of vertices
E	Set of edges
$\mathbf{S}$	Matrix/submatrix of shares
$\mathbf{v}$	Unit vectors
ι	Impact modifiers
ϕ	Cosine similarity modifier

12.4 Capital & Markets

$s \in \hat{S} \subset S$	Shares
Ӿ	Academic capital
ΔӾ	Return
μ	Expected returns
σ	Volatility
γ	Risk asymmetry

ϵ	Efficiency
$\bar{s}_p, \bar{s}_r \in \bar{S}$	Fair market prices
ψ	Risk premium
ν	Utility
α	Excess returns (outperformance)
β	Market sensitivity
$d_{1,2,3,4} \in D_{1,2,3,4}$	Academic domain, department, discipline, or direction
ARC	Academic returns to capital ratio
HHI	Herfindahl–Hirschman Index
Gini	Gini Index
$\pi \in \Pi$	Portfolio

References

Adil E. Shamoo and David B. Resnik. *Responsible Conduct of Research*. Oxford University Press, 2009.

Henry Sauermann and Carolin Haeussler. Authorship and contribution disclosures. *Science Advances*, 3(11), 2017.

David Riesenberg and George D. Lundberg. The order of authorship: who's on first? *JAMA*, 264(14):1857, 1990.

Teja Tscharntke, Michael E. Hochberg, Tatyana A. Rand, Vincent H. Resh, and Jochen Krauss. Author sequence and credit for contributions in multiauthored publications. *PLoS Biology*, 5(1):e18, 2007.

Liran Einav and Leeat Yariv. Alphabetical discrimination in academic publishing. *Quarterly Journal of Economics*, 121(1):1–41, 2006.

Ludo Waltman. An empirical analysis of the use of alphabetical authorship in scientific publishing. *Journal of Informetrics*, 6(4):700–711, 2012.

David M. Bennett and David M. Taylor. Authorship attribution: survey of guidelines in biomedical journals. *Journal of Medical Ethics*, 29(6):345–348, 2003.

L. Allen, J. Scott, A. Brand, M. Hlava, and M. Altman. Credit where credit is due: The credit taxonomy. *Learned Publishing*, 27(2):123–127, 2014. doi:10.1087/20140211.

H. W. Shen and A. L. Barabási. Collective credit allocation in science. *Proceedings of the National Academy of Sciences*, 111(34):12325–12330, 2014. doi:10.1073/pnas.1401992111.

N. T. Hagen. Harmonic allocation of authorship credit: Source-level correction of bibliometric bias assures accurate publication and citation analysis. *PLoS ONE*, 3(12):e4021, 2008. doi:10.1371/journal.pone.0004021.

Leo Egghe. Mathematical theory of the h- and g-index in case of fractional counting of authorship. *Journal of the American Society for Information Science and Technology*, 2008.

Antonio Perianes-Rodriguez, Ludo Waltman, and Nees Jan van Eck. Constructing bibliometric networks: A comparison between full and fractional counting. *Journal of Informetrics*, 10(4):1178–1195, 2016.

Chun-Ting Zhang. A proposal for calculating weighted citations based on author rank. *EMBO reports*, 10 (5):416, 2009.

Gary S. Howard, Deborah A. Cole, and Scott E. Maxwell. A method for assigning co-authorship credit. *Scientometrics*, 72(3):523–538, 2007.

M. M. Kessler. Bibliographic coupling between scientific papers. *American Documentation*, 14(1):10–25, 1963.

Henry Small. Co-citation in the scientific literature: A new measure of the relationship between two documents. *Journal of the American Society for Information Science*, 24(4):265–269, 1973.

Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The pagerank citation ranking: Bringing order to the web. *Stanford InfoLab Technical Report*, 1999.

P. Chen, H. Xie, S. Maslov, and S. Redner. Finding scientific gems with google's pagerank algorithm. *Journal of Informetrics*, 1(1):8–15, 2007. doi:10.1016/j.joi.2006.06.001.

D. Walker, H. Xie, K.-K. Yan, and S. Maslov. Ranking scientific publications using a model of network traffic. *Journal of Statistical Mechanics: Theory and Experiment*, 2007(06):P06010, 2007.

Ludo Waltman, Nees Jan van Eck, Thed N. van Leeuwen, Martijn S. Visser, and Anthony F. J. van Raan. Towards a new crown indicator: Some theoretical considerations. *Journal of Informetrics*, 5(1):37–47, 2011.

Alan L. Porter and Ismael Rafols. Measuring researcher interdisciplinarity. *Scientometrics*, 72(1):117–147, 2009.

Caroline S. et al. Wagner. Approaches to understanding and measuring interdisciplinary scientific research (idr). *Journal of Informetrics*, 5(1):14–26, 2011.

Dashun Wang, Chaoming Song, and Albert-László Barabási. Quantifying long-term scientific impact. *Science*, 342(6154):127–132, 2013.

Pietro D. B. et al. Parolo. Attention decay in science. *PNAS*, 112(20):6426–6431, 2015.

Ludo Waltman and Nees Jan van Eck. A systematic empirical comparison of different approaches for normalizing citation impact indicators. *Journal of Informetrics*, 7(4):833–849, 2013.